

Notes: Hamilton (JPE, 1983)

Oil and the Macroeconomy since World War II

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1 Big picture

- **Question.** Did postwar increases in crude oil prices merely coincide with U.S. recessions, or did oil shocks play a causal role in postwar business cycles?
- **Setting.** The paper studies U.S. macroeconomic time series after World War II, with particular attention to the pre-OPEC period 1948–1972 and the oil-shock period after 1973.
- **Central fact.** All but one U.S. recession after World War II was preceded, usually by about three quarters, by a sharp increase in crude oil prices.
- **Core challenge.** A correlation between oil prices and recessions has three possible interpretations:
 - it could be a historical coincidence;
 - oil prices could be endogenous, responding to the same macroeconomic forces that cause recessions;
 - oil-price increases could themselves be a contributing cause of recessions.
- **Main empirical strategy.** Hamilton combines historical evidence on the institutional sources of oil-price changes with Granger-causality tests. The key question is whether macro variables predict oil prices, and whether oil prices predict future output and unemployment.
- **Main result.** Oil price changes forecast later real GNP growth and unemployment, but the major macro variables generally do not forecast oil price changes. This supports the view that oil shocks were not just endogenous reflections of the business cycle.
- **Important nuance.** The paper does *not* claim that oil shocks are necessary or sufficient for recessions. The 1960 recession occurred without a major oil shock, and the 1970–1971 oil price increases were not sufficient by themselves to create the same pattern.
- **Economic takeaway.** Supply-side energy shocks mattered before the 1970s as well. The paper argues that postwar macroeconomic history should combine demand-side explanations with an explicit role for energy-price disturbances.

2 Data and measurement (key objects)

2.1 Observed macroeconomic data

- **Oil price variable.** The main oil variable is the change in the log crude oil wholesale price index, measured using end-of-quarter values.
- **Main sample.** The core statistical tests focus on 1948–1972. This is important because the post-1973 OPEC period is used partly as motivation rather than as the source of the hypothesis.

- **Post-1973 comparison.** The paper also examines 1973–1980 to ask whether the pre-1973 relation between oil and output resembles the post-OPEC relation.
- **Macro variables from Sims’s system.** Hamilton uses a compact six-variable macroeconomic system inspired by Sims (1980b):
 - real GNP growth;
 - change in unemployment;
 - domestic price inflation;
 - wage inflation;
 - change in money growth;
 - import price inflation.
- **Alternative indicators.** The paper also examines nominal GNP, inventories-to-sales, the BEA leading indicator index, capacity utilization, strike activity, wholesale prices, commodity price indexes, coal prices, BAA bond yields, and the Dow-Jones Industrial Average.

2.2 Historical oil-price episodes

- The paper lists major crude oil price episodes from 1947–1981 and links them to historically specific energy-sector events.
- Examples include:
 - 1952–1953: Iranian nationalization, strikes by oil/coal/steel workers, and the import posture of the Texas Railroad Commission;
 - 1956–1957: the Suez crisis;
 - 1969: secular decline in U.S. reserves and oil-worker strikes;
 - 1970: rupture of the trans-Arabian pipeline, Libyan production cutbacks, and coal-price increases;
 - 1973–1974: stagnating U.S. production and the OPEC embargo;
 - 1978–1979: the Iranian revolution;
 - 1980–1981: the Iran-Iraq war and removal of U.S. price controls.
- **Institutional mechanism before OPEC.** In the pre-1973 period, the Texas Railroad Commission and similar agencies regulated production allowables. Much of the cyclical variation in petroleum demand showed up in quantities rather than prices.
- **Implication.** Because regulators often did not accommodate sudden disruptions in supply, large discrete price increases frequently reflected sector-specific disruptions rather than ordinary aggregate-demand pressure.

2.3 Descriptive facts

- Real GNP growth fell from about **4.0 percent** during 1960–1972 to about **2.4 percent** during 1973–1981.
- Inflation rose from about **3.1 percent** during 1960–1972 to about **7.6 percent** during 1973–1981.
- Unemployment averaged about **6.7 percent** during 1973–1981, higher than in any year between 1948 and 1972 except 1958.
- The real price of energy had roughly doubled relative to its 1972 level by the end of the post-OPEC period discussed in the paper.
- Figure 1 visually motivates the analysis: oil price spikes tend to appear shortly before recession shading.

2.4 Unobservables and timing

- **Timing.** The key lag is roughly three to four quarters from an oil-price increase to a downturn in output.
- **Potential omitted factor.** A hidden macroeconomic shock could cause both oil prices and recession. This is the central alternative explanation the paper tries to rule out.
- **Nonstationarity concern.** The oil market changed substantially after 1973, and many macro series have trends. Hamilton therefore focuses on pre-1973 tests and filters variables to remove nonstationarity.
- **Data-mining concern.** Because the post-1973 experience suggests the hypothesis, the pre-1973 sample is useful as a separate test of whether the same oil-output pattern existed before OPEC.

3 Models and empirical specifications (all equations + interpretation)

3.1 Three hypotheses about the oil-recession correlation

Hamilton organizes the argument around three interpretations:

1. **Coincidence.** Oil price increases just happened to occur near recessions.
2. **Endogenous oil prices.** Some third macroeconomic force caused both oil price increases and recessions.

3. **Causal oil shocks.** Exogenous oil price increases contributed to the timing, magnitude, or duration of at least some recessions.

Interpretation. The paper does not try to prove that oil shocks are the only cause of recessions. It asks whether the evidence is strong enough to reject the first two interpretations and take the third seriously.

3.2 Basic Granger-causality regression

The central statistical exercise is based on regressions of the form

$$z_t = a_0 + a_1 z_{t-1} + \cdots + a_q z_{t-q} + b_1 x_{t-1} + \cdots + b_q x_{t-q} + e_t. \quad (1)$$

The null hypothesis is

$$H_0 : b_1 = b_2 = \cdots = b_q = 0. \quad (2)$$

- If x_t is a macro variable and z_t is oil-price growth, rejection means the macro variable helps predict oil prices.
- If x_t is oil-price growth and z_t is a macro variable, rejection means oil prices help predict the macro variable.
- Hamilton estimates these tests mainly with $q = 4$ and $q = 8$ quarterly lags.

Interpretation. Oil prices are more plausibly exogenous if macro variables do *not* predict oil prices, while oil prices do predict future output and unemployment.

3.3 Variables in the Sims-style macro system

Hamilton defines the main variables roughly as:

$$y_t = \log(GNP72_t / GNP72_{t-1}), \quad (3)$$

$$u_t = \Delta UN_t, \quad (4)$$

$$p_t = \log(PYNDOMNF_t / PYNDOMNF_{t-1}), \quad (5)$$

$$w_t = \log(HRWSSNF_t / HRWSSNF_{t-1}), \quad (6)$$

$$m_t = \log(MONEYNS_t / MONEYNS_{t-1}) - \log(MONEYNS_{t-1} / MONEYNS_{t-2}), \quad (7)$$

$$p_t^m = \log(PM_t / PM_{t-1}), \quad (8)$$

$$o_t = \log(OIL_t / OIL_{t-1}). \quad (9)$$

Here y_t is real GNP growth, u_t is the change in unemployment, p_t is domestic price inflation, w_t is wage inflation, m_t is the change in money growth, p_t^m is import-price inflation, and o_t is oil-price inflation.

Interpretation. The goal is to test whether oil shocks still matter when the main output, labor-market, price, wage, monetary, and import-price variables are taken into account.

3.4 Anticipated versus unanticipated components

When a variable such as import prices appears to predict oil prices, Hamilton asks whether this is because it proxies for general macroeconomic conditions. The first step is to predict import-price inflation using lagged macro variables:

$$p_t^m = \alpha + X'_{t-1}a_1 + X'_{t-2}a_2 + \cdots + X'_{t-8}a_8 + \varepsilon_t, \quad (10)$$

where

$$X_t = (y_t, u_t, p_t, w_t, m_t, p_t^m)'. \quad (11)$$

Then decompose import-price inflation into:

$$p_t^m = \hat{p}_t^m + (p_t^m - \hat{p}_t^m). \quad (12)$$

The oil-price regression includes both the anticipated and unanticipated components:

$$o_t = c_0 + \sum_{i=1}^8 b_i o_{t-i} + \sum_{i=1}^8 c_i \hat{p}_{t-i}^m + \sum_{i=1}^8 d_i (p_{t-i}^m - \hat{p}_{t-i}^m) + \eta_t. \quad (13)$$

Interpretation. If only the anticipated part predicts oil, then import prices may just be proxying the business cycle. If the unanticipated part matters, then the predictive content is more sector-specific and less consistent with a general macro omitted-variable story.

3.5 Multivariate tests

Hamilton also runs multivariate versions of the tests. For example, to test whether oil predicts real GNP growth conditional on the Sims macro system, the regression includes lags of real GNP, unemployment, prices, wages, money, import prices, and oil:

$$y_t = \alpha + \sum_{i=1}^q A_i X_{t-i} + \sum_{i=1}^q b_i o_{t-i} + e_t. \quad (14)$$

The null is again

$$H_0 : b_1 = \dots = b_q = 0. \quad (15)$$

Interpretation. This asks whether oil prices contain information about future output beyond what is already contained in the main macroeconomic variables.

3.6 Specification analysis: nonnormal oil shocks

Because oil price changes contain large spikes, Hamilton emphasizes that they are not well approximated by a normal distribution. He uses a Lomnicki-style test statistic for excess kurtosis:

$$L = \frac{\sqrt{T} (m_4/m_2^2 - 3)}{\sqrt{24}}, \quad (16)$$

where m_2 and m_4 are sample second and fourth moments of the residuals.

Interpretation. The oil-price residuals are highly leptokurtic. This matters because standard small-sample F tests may behave poorly when the dependent variable has extreme jumps.

3.7 Sims-Geweke alternative causality test

To avoid relying too heavily on regressions with oil prices as the dependent variable, Hamilton also uses an alternative test where oil-price leads appear as regressors:

$$y_t = \alpha + \sum_{i=1}^8 a_i y_{t-i} + \sum_{i=0}^8 b_i o_{t-i} + \sum_{i=1}^s c_i o_{t+i} + \mu_t. \quad (17)$$

The null that output does not Granger-cause oil is written as

$$H_0 : c_1 = \dots = c_s = 0. \quad (18)$$

Interpretation. If future oil prices help explain current output after controlling for current and lagged oil prices, then output may help predict oil. Hamilton generally does not find this for the Sims macro variables.

3.8 Structural stability and functional form

The paper estimates the oil-output relation:

$$y_t = a_0 + a_1 y_{t-1} + \dots + a_4 y_{t-4} + b_1 o_{t-1} + \dots + b_4 o_{t-4} + u_t. \quad (19)$$

- The pre-1973 relation differs from the post-1973 relation in a strict linear-coefficient sense.
- But the dynamic pattern is similar: oil price increases are followed by slower output growth after about three to four quarters, with recovery after about six to seven quarters.
- The post-1973 coefficients tend to be smaller, which Hamilton links to inflation, policy responses, and changes in the oil-price regime.

Interpretation. The relation should not be treated as an immutable structural law. The same economic shock can have different reduced-form coefficients under different policy regimes and inflation environments.

4 Identification and instruments (what makes the estimates credible)

4.1 What has to be identified

The paper needs to distinguish three possibilities:

- oil prices and recessions are unrelated except by chance;
- oil prices are endogenous to the same macro forces that cause recessions;
- oil price increases are at least partly exogenous and contribute to downturns.

Key identification problem. Oil prices are not randomly assigned. The credibility of the causal interpretation depends on showing that oil-price changes were not simply responses to the business cycle.

4.2 Historical identification: why oil shocks look plausibly exogenous

- Hamilton emphasizes the regulatory structure of the pre-OPEC U.S. oil market.
- Regulatory agencies adjusted quantities through production allowables, which filtered out much of the normal cyclical component of petroleum demand.
- Large price movements were instead associated with discrete disruptions or sector-specific events such as wars, strikes, foreign nationalizations, pipeline disruptions, or resource constraints.

Interpretation. This historical argument is what gives the statistical tests economic meaning. The regressions alone show predictive ordering, but the institutional evidence explains why oil-price movements may be plausibly exogenous.

4.3 Statistical identification: what predicts what

- If macro variables predicted oil prices, that would support the endogenous-oil-price interpretation.
- If oil prices predicted macro variables, while macro variables did not predict oil prices, that would support the oil-shock interpretation.
- Hamilton finds the second pattern: oil predicts output and unemployment, but output, unemployment, wages, prices, money, interest rates, and stock prices generally do not predict oil.

Interpretation. This pattern is not a proof of causality in the modern experimental sense, but it makes the pure reverse-causality story less convincing.

4.4 Why alternative indicators matter

Hamilton checks whether other variables could explain the oil-output relation:

- **Output indicators:** nominal GNP, inventories-to-sales, BEA leading indicators, and capacity utilization.
- **Price indicators:** wholesale prices and several commodity price indexes.
- **Financial indicators:** interest rates and stock prices.
- **Energy-related variables:** strike activity and coal prices.

Interpretation. The broad set of checks is designed to rule out the idea that oil is just a proxy for overheating, commodity inflation, credit tightening, or financial-market expectations.

4.5 Main limitations

- Granger causality is predictive, not structural causality.
- Oil-price changes are highly nonnormal, which complicates inference.
- The post-1973 oil-price regime differs from the pre-1973 regime, so reduced-form coefficients need not be stable.
- The paper does not estimate a fully specified structural model of production, energy use, monetary policy, and inflation.
- Some oil-price movements may reflect broader energy-market shocks, especially when coal prices and strike activity move together with oil.

4.6 Relation to the literature

- **Granger and Sims time-series methods.** The paper uses the then-new time-series framework for testing predictive ordering in macroeconomic data.
- **Monetarist and demand-side business-cycle views.** Hamilton challenges explanations that rely only on aggregate demand or monetary variables.
- **Supply-shock macroeconomics.** The paper provides early evidence that oil-price disturbances are important macroeconomic supply shocks.
- **Later oil-shock literature.** The paper became a foundation for research on oil shocks, nonlinear energy-price effects, monetary-policy responses, and the macroeconomic consequences of commodity-price disruptions.

5 Tables and figures

5.1 Figure 1: crude oil prices and U.S. recessions

Read:

- Figure 1 plots quarter-to-quarter percent changes in crude oil wholesale prices against U.S. recession shading from 1947–1975.
- The main visual fact is striking: oil price increases appear shortly before almost every postwar recession.
- The exception is the 1960–1961 recession.
- The figure motivates the paper, but Hamilton is careful not to treat the visual correlation as proof of causality.
- The rest of the paper asks whether this pattern survives formal statistical and historical scrutiny.

5.2 Table 1: principal causes of crude oil price increases

Read:

- Table 1 is the historical foundation of the identification argument.
- The listed events are mostly petroleum-sector or geopolitical events rather than ordinary U.S. aggregate-demand shocks.
- The table helps explain why Hamilton treats major oil-price movements as plausibly exogenous to the U.S. macroeconomy.

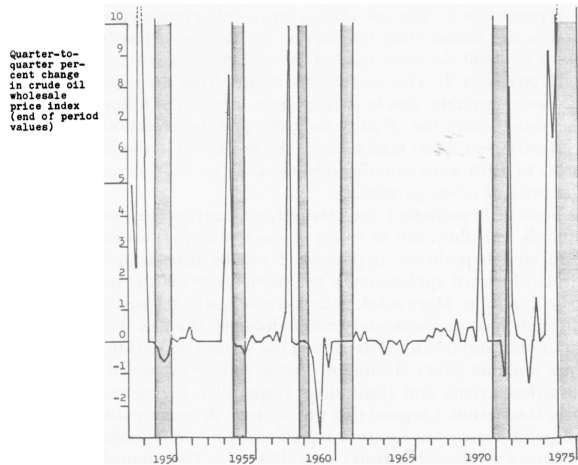


FIG. 1.—Changes in crude oil prices (solid lines) and U.S. recessions (shaded areas), 1947–75.

TABLE 1
PRINCIPAL CAUSES OF CRUDE OIL PRICE INCREASES, 1947–81

Oil Price Episode	Principal Factors
1947–48	Previous investment in production and transportation capacity inadequate to meet postwar needs; decreased coal production resulting from shorter work week; European reconstruction
1952–53	Iranian nationalization; strikes by oil, coal, and steel workers; import posture of Texas Railroad Commission
1956–57	Suez crisis
1969	Secular decline in U.S. reserves; strikes by oil workers
1969	Rupture of trans-Arabian pipeline; Libyan production cutbacks; coal price increases (strikes by coal workers; increased coal exports; environmental legislation)
1973–74	Stagnating U.S. production; OPEC embargo
1978–79	Iranian revolution
1980–81	Iran-Iraq war; removal of U.S. price controls

SOURCE.—Hamilton (1982a).

- This historical evidence is especially important because pure Granger-causality tests cannot by themselves establish structural causality.

5.3 Table 2: bivariate Granger-causality tests

Read:

- Real GNP growth does not predict oil prices: the p -values are about 0.68 with both four and eight lags.
- Oil prices strongly predict real GNP growth: the four-lag test gives $F(4, 86) = 5.55$ with $p = 0.0005$, and the eight-lag test gives $F(8, 74) = 3.28$ with $p = 0.003$.
- Unemployment also shows the asymmetric pattern: unemployment does not robustly predict oil, but oil predicts unemployment with $p = 0.007$ and $p = 0.005$ in the four- and eight-lag tests.
- Domestic prices, wages, and money growth do not provide strong evidence of predicting oil.
- Import prices are the main exception: with eight lags, import prices appear to predict oil. Hamilton investigates this further rather than ignoring it.

5.4 Table 3: multivariate Granger tests

Read:

- The six-variable Sims system collectively does not strongly predict oil prices at the 5 percent level, though the result is close.
- Conditional on the full Sims system, oil still predicts real GNP growth: $F(4, 66) = 5.03$ with $p = 0.0003$.

TABLE 2
BIVARIATE GRANGER-CAUSALITY TESTS FOR SIX VARIABLES IN
SIMS'S (1980b) SYSTEM VERSUS CHANGES IN OIL PRICES

Null Hypothesis	$q = 4, T = 95$ ($t = 1949:2-1972:4$)		$q = 8, T = 91$ ($t = 1950:2-1972:4$)	
	$F(4,86)$	(p -value)	$F(8,74)$	(p -value)
Real GNP: $y_t = \log(\text{GNP72}/\text{GNP72}_{t-1})$:				
$H_1: y \not\rightarrow o$	.58	(.68)	.71	(.68)
$H_2: o \not\rightarrow y$	5.55	(.0005)	3.28	(.003)
Unemployment: $u_t = \text{RUNS}_t$:				
$H_3: u \not\rightarrow o$	2.06	(.09)	1.48	(.18)
$H_4: o \not\rightarrow u$	3.79	(.007)	3.08	(.005)
U.S. prices: $p_t = \log(\text{PYNDOMNF}/\text{PYNDOMNF}_{t-1})$:				
$H_5: p \not\rightarrow o$	2.01	(.10)	1.54	(.16)
$H_6: o \not\rightarrow p$	.40	(.81)	.31	(.96)
Wages: $w_t = \log(\text{JRWSSNF}/\text{JRWSSNF}_{t-1})$:				
$H_7: w \not\rightarrow o$	1.92	(.11)	1.46	(.19)
$H_8: o \not\rightarrow w$	1.28	(.28)	1.32	(.25)
Money: $m_t = \log(\text{MONEYNS}/\text{MONEYNS}_{t-4})$ - $\log(\text{MONEYNS}_{t-1}/\text{MONEYNS}_{t-5})$:				
$H_9: m \not\rightarrow o$	.42	(.79)	.49	(.86)
$H_{10}: o \not\rightarrow m$	1.10	(.36)	2.22	(.04)
Import prices: $p_t^m = \log(\text{PM}_t/\text{PM}_{t-1})$:				
$H_{11}: p^m \not\rightarrow o$	.55	(.70)	3.21	(.003)
$H_{12}: o \not\rightarrow p^m$	.22	(.93)	.26	(.98)
$H_{13}: p^m \not\rightarrow y$	.92	(.46)	1.01	(.44)

TABLE 3
MULTIVARIATE GRANGER-CAUSALITY TESTS

$H_1: y, u, p, w, m, p^m \not\rightarrow o$ $t = 1949:2-1972:4, 4$ lags each variable $F(4,66) = 1.61, p = .065$
$H_2: o \not\rightarrow y$ given past y, u, p, w, m, p^m $t = 1949:2-1972:4, 4$ lags each variable $F(4,66) = 6.03, p = .0003$
$H_3: o \not\rightarrow u$ given past y, u, p, w, m, p^m $t = 1949:2-1972:4, 4$ lags each variable $F(4,66) = 3.99, p = .006$
$H_4: o \not\rightarrow y$ given past y, p^m $t = 1950:2-1972:4, 8$ lags each variable $F(8,66) = 3.45, p = .002$
$H_5: o \not\rightarrow y$ given past y, s $t = 1950:2-1972:4, 8$ lags each variable $F(8,66) = 3.18, p = .004$
$H_6: o \not\rightarrow y$ given past y, p' $t = 1950:2-1972:4, 8$ lags each variable $F(8,66) = 3.34, p = .003$
$H_7: o \not\rightarrow y$ given past y, p^m, s, p' $t = 1949:2-1972:4, 4$ lags each variable $F(4,74) = 3.39, p = .013$

- Conditional on the same variables, oil also predicts unemployment: $F(4,66) = 3.99$ with $p = 0.006$.
- Oil remains statistically informative about output after controlling for import prices, strike activity, coal prices, or combinations of these controls.
- The table is the paper's central statistical evidence against the claim that oil simply proxies for standard macro variables.

5.5 Table 4: alternative indicators of output, prices, and finance

Read:

- The alternative output indicators generally fail to predict oil prices: nominal GNP, inventories-to-sales, leading indicators, and capacity utilization are not significant predictors.
- Most non-energy price indexes also fail to predict oil prices: wholesale prices, nonferrous metals, iron and steel, farm products, and lumber products do not mimic the oil pattern.
- Coal prices are the strongest price-side predictor of oil, with very small p -values, especially in the eight-lag test.
- Strike activity matters in the four-lag test, consistent with the historical role of oil, coal, and steel strikes in several oil-price episodes.
- Financial variables do not support a hidden-recession story: BAA yields and the Dow-Jones Industrial Average do not forecast oil price changes in a way that explains the oil-output relation.

TABLE 4
MARGINAL SIGNIFICANCE LEVELS FOR REJECTION OF NULL HYPOTHESIS THAT x DOES NOT GRANGER-CAUSE OIL PRICES, FOR x ALTERNATIVE INDICATORS OF OUTPUT, PRICES, AND FINANCE, 1948–72

x	Filter for Stationarity	$F(4,86)$ p -value	$F(8,74)$ p -value
Output:			
1. Nominal GNP	$(1 - L)(1 - L^3)\log$	(.60)	(.50)
2. Inventories/sales	$(1 - L)$	(.58)	(.75)
3. BEA Index of Leading Indicators	$(1 - L)$	(.56)	(.61)
4. FRB Index of Capacity Utilization	1	(.27)	(.33)
5. Strikes	1	(.04)*	(.13)
Prices:			
6. Relative oil prices	1	(.13)	(.18)
7. Wholesale prices	$(1 - L)\log$	(.76)	(.88)
8. Nonferrous metals	$(1 - L)\log$	(.87)	(.45)
9. Iron and steel	$(1 - L)\log$	(.13)	(.12)
10. Farm products	$(1 - L)\log$	(.38)	(.55)
11. Lumber products	$(1 - L)\log$	(.90)	(.42)
12. Coal	$(1 - L)\log$	(.003)**	(.004)**
Finance:			
13. BAA Bond Yields	$(1 - L)$	(.10)	(.55)
14. Dow-Jones Industrial Average	$(1 - L)\log$	(.68)	(.44)

NOTE.—BEA = Bureau of Economic Analysis; FRB = Federal Reserve Board; BAA = a type of rating given to bonds that reflects a company's financial status.

* Statistically significant at 5 percent level.

** Statistically significant at 1 percent level.

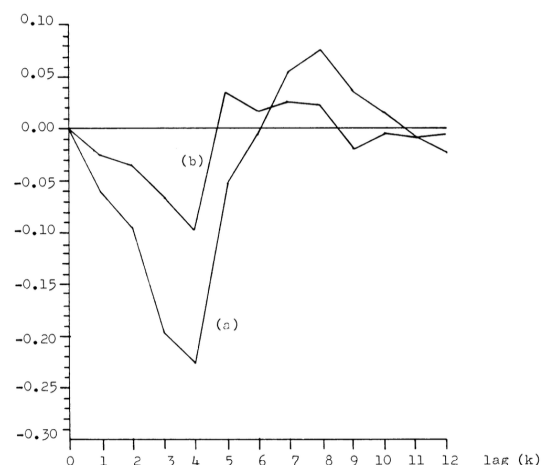


FIG. 2.—“Final form” relation between changes in real GNP and previous changes in oil prices as estimated for (a) 1948–72; (b) 1973–80.

5.6 Figure 2: final-form relation between oil prices and output

Read:

- Figure 2 plots the implied dynamic response of real GNP growth to oil-price changes.
- In the pre-1973 sample, an oil-price increase is followed by slower output growth after roughly three to four quarters.
- The response then begins to reverse after roughly six to seven quarters.
- The post-1973 pattern has similar timing but different coefficient magnitudes.
- The figure shows why Hamilton emphasizes timing: the oil shock is not contemporaneous with the recession; it tends to lead the downturn.

5.7 Table 5: structural stability of the oil-output relation

Read:

- Table 5 estimates the relation between real GNP growth and lagged oil-price changes over different sample periods.
- For 1949:2–1972:4, lagged oil prices significantly predict real GNP growth, with $F(4, 86) = 5.55$ and $p = 0.0005$.
- For 1973:1–1980:3, oil prices also significantly predict real GNP growth, with $F(4, 22) = 5.71$ and $p = 0.003$.
- A strict linear model is not fully stable across periods, which is not surprising because the oil market and inflation environment changed after 1973.

TABLE 5
REGRESSION COEFFICIENTS RELATING y_t TO o_t FOR INDICATED SAMPLE PERIODS, WITH
 F -TEST THAT COEFFICIENTS ON LAGGED o_t ARE ALL ZERO

COEF- FICIENT	SAMPLE PERIOD*			
	1949:2–1972:4 (1)	1973:1–1980:3 (2)	1948:2–1972:4 (3)	1948:2–1980:3 (4)
Constant	.011 (.0016)	.023 (.0050)	.0088 (.0017)	.010 (.0016)
y_{t-1}	.19 (.097)	-.20 (.18)	.30 (.10)	.21 (.088)
y_{t-2}	.15 (.097)	-.064 (.20)	.10 (.11)	.16 (.092)
y_{t-3}	-.049 (.097)	-.088 (.20)	-.098 (.10)	-.081 (.092)
y_{t-4}	-.28 (.092)	-.34 (.19)	-.26 (.10)	-.25 (.088)
o_{t-1}	-.061 (.056)	-.023 (.028)	-.041 (.061)	.004 (.023)
o_{t-2}	-.082 (.056)	-.038 (.028)	.019 (.036)	-.007 (.021)
o_{t-3}	-.170 (.057)	-.078 (.030)	-.057 (.035)	-.050 (.021)
o_{t-4}	-.177 (.059)	-.115 (.033)	-.047 (.035)	-.062 (.021)
F -statistic	$F(4,86) =$ 5.55	$F(4,22) =$ 5.71	$F(4,90) =$ 1.51	$F(4,121) =$ 5.28
p -value	.0005	.003	.21	.0007

NOTE.— y_t = quarterly changes in log of real GNP; o_t = quarterly changes in nominal end-of-period crude oil prices. SEs of coefficients are in parentheses.
* Bounds on t in regression 7.

- The economic conclusion is more modest: oil shocks matter, but the reduced-form coefficients are regime-dependent.

6 Short summary

- **Method.** The paper combines historical analysis of oil-market disruptions with Granger-causality tests in a compact macro time-series framework.
- **Identification.** Oil shocks are treated as plausibly exogenous because the main price episodes are linked to sector-specific or geopolitical disruptions, and because standard macro variables generally do not forecast oil prices.
- **Empirical result.** Oil-price increases predict lower future real GNP growth and higher unemployment, typically with a lag of about three to four quarters.
- **Robustness.** Alternative output, price, and financial indicators do not explain away the oil-output relationship. Coal prices and strikes matter, but they are interpreted as energy-sector or sector-specific shocks rather than broad macro proxies.
- **Main conclusion.** Oil shocks were a contributing factor in at least some U.S. recessions before 1973, and energy-price increases may explain much of the difficult post-OPEC macroeconomic experience.

7 Conclusion

7.1 Core conclusion

- The paper rejects the view that the oil-recession correlation is merely coincidental.
- It also finds little evidence that oil prices were simply endogenous responses to ordinary U.S. macroeconomic conditions.
- The evidence supports the view that oil-price increases helped shape the timing, magnitude, or duration of some postwar U.S. recessions.

7.2 Economic intuition

- Oil is a key production input and consumption good, so a large price increase can reduce real income, raise costs, disrupt production, and change spending patterns.
- The effect is delayed because firms, households, and policy makers adjust gradually.
- If monetary policy reacts to oil-induced inflation by tightening, the total macroeconomic effect can be larger than the direct supply shock alone.
- But because policy regimes and inflation environments change, the same oil-price movement need not have the same effect in every historical period.

7.3 Takeaway

Hamilton's paper is important because it made oil shocks central to empirical macroeconomics. Its key contribution is not just showing that oil prices lead recessions; it is showing that this pattern existed before OPEC and is difficult to explain using standard macro predictors. The paper therefore pushes macroeconomic analysis toward a synthesis: postwar recessions cannot be understood solely through demand, money, or financial variables; energy-sector supply disturbances also belong in the story.